

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Winter Examination – 2022****Course: B. Tech.****Branch : Electrical Engineering****Semester :V****Subject Code & Name: BTEEC502, Microprocessor and Microcontroller****Marks: 60****Date:31/01/23****Duration: 3 Hr.****Instructions to the Students:****All the questions are compulsory.****The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.****Use of non-programmable scientific calculators is allowed.****Assume suitable data wherever necessary and mention it clearly.**

		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		12
A)	Explain different types of addressing modes of 8085 microprocessor.	CO1	6
B)	Explain traffic light controller in 8085 microprocessor.	CO4	6
C)	What is an interrupt? Explain a different type of interrupt occurs in 8085?	CO3	6
Q.2	Solve Any Two of the following.		12
A)	Draw and Explain timing dia. of opcode fetch operation in 8085 microprocessors.	CO1	6
B)	Discuss different types of Data transfer techniques in 8085.Explain Synchronous and Asynchronous Data Transfer Method.	CO2	6
C)	Explain different types of addressing modes in 8051 microcontroller.	CO5	6
Q. 3	Solve Any Two of the following.		12
A)	Draw and explain block dia. of 8051 microcontroller.	CO5	6
B)	Explain what is A. Memory mapped I/O schemes. B.I/O mapped I/O schemes	CO2	6
C)	Explain power saving mode in 8051.	CO6	6
Q.4	Solve Any Two of the following.		12
A)	Draw Architecture dia. of 8085 microprocessor.	CO1	6
B)	Explain interfacing of ADC 0800 converter.	CO4	6
C)	Write a programme for addition and subtraction using 8085 instruction set.	CO1	6
Q. 5	Solve Any Two of the following.		12
A)	Explain DMA controller.	CO3	6
B)	Write short note on a. instruction cycles b. machine cycle c. T-state.	CO1	6
C)	Explain TCON and TMOD register in 8051 microcontroller.	CO6	6

***** End *****